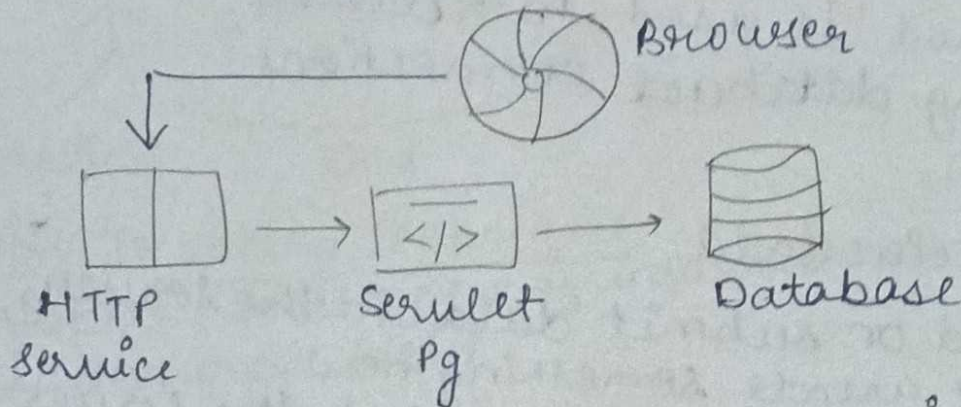


Unit-5

* Servlets with its lifecycle

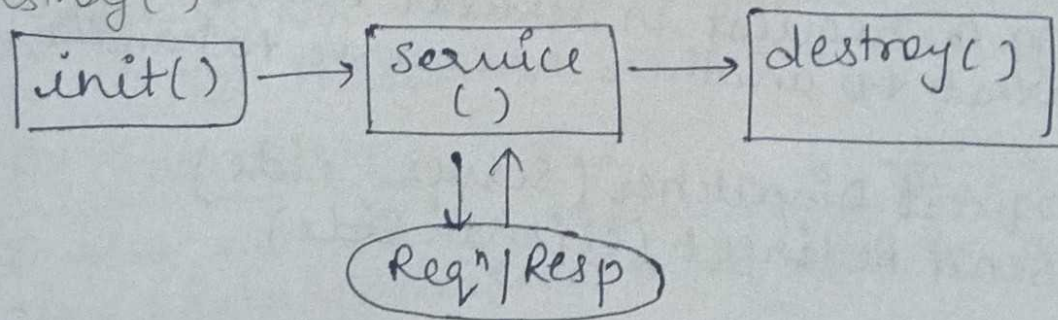
- ↳ Servlets are the Java program that run on the Java-enabled web server or application server.
- ↳ They are used to handle the request obtained from the web-server, process the reqⁿ, produce the response and then send a response to the web server.
- ↳ Servlets work on the server side.



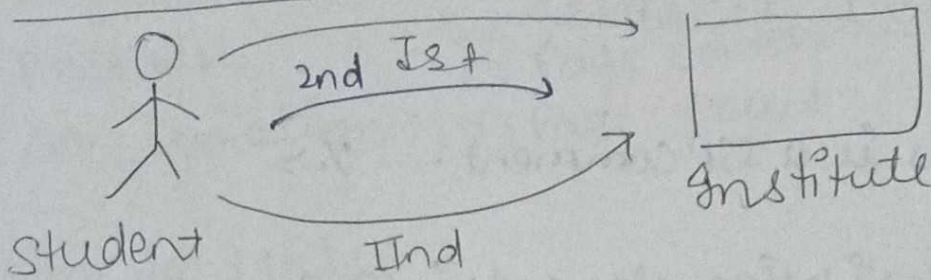
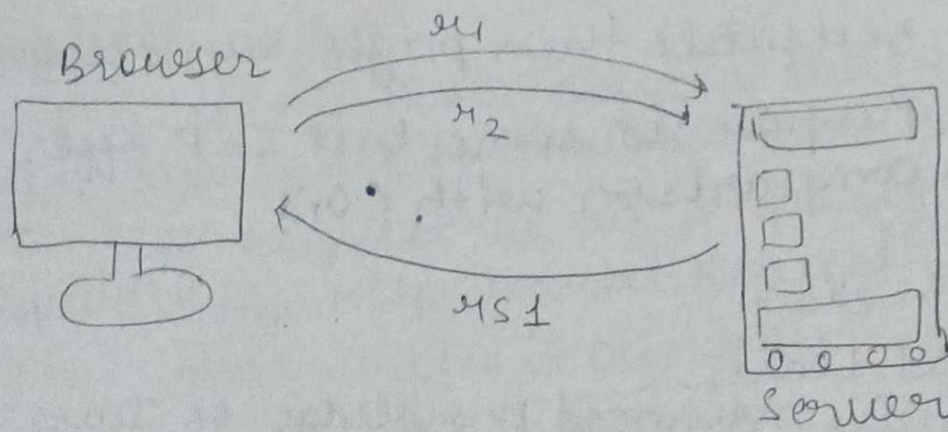
- ↳ Servlets are help to create dynamic web content and it is more efficient than older CGI scripts.

Lifecycle

- `init()`
- `service()`
- `destroy()`



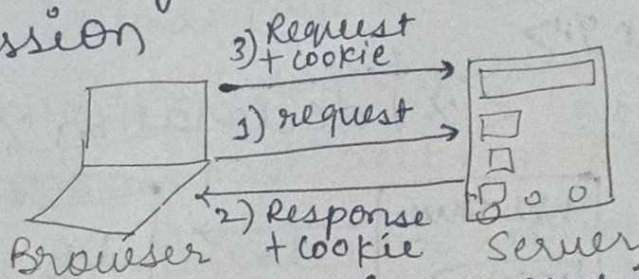
- ↳ Init:- `init()` Method is called only once
 - ↳ created only when the servlet created
 - ↳ The `init()` method must complete successfully before the servlet can receive any request and response
- ↳ when the servlet is first requested, the container creates an instances of servlet class.



Session Tracking Technique

- 1) Cookies (small piece of inform stored on the client)
- 2) Hidden form field
- 3) URL Rewriting
- 4) HTTP session

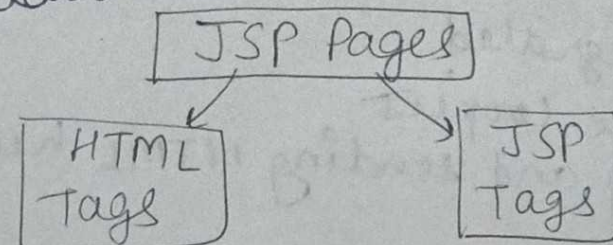
↳ Cookies



↳ Types → ① Non-Persistent cookie (valid for single session)
 → ② Persistent (valid for multiple session)

JSP (Java Server Page) *

- ↳ JSP is a serverside technology that is used for creating web application. It is used to create dynamic web content.
- ↳ It supports both scripting and element based dynamic content.
- ↳ JSP is an advanced version of servlet technology.



Implicit object:

→ are a set of Java objects that the JSP container makes available to developers on each page.

→ The JSP request is an implicit object which is provided by HTTP servlet Request.

Total Nine implicit objects as follows:

1) Request

2) Response

3) Configuration

4) application

5) Session

6) Page Context

7) Page object

8) Exception

9) out

Standard Actions:

In JSP, standard actions are special JSP tags that are used to perform common tasks like forwarding request, including files or interacting with JavaBeans.

1) <jsp:include> To include another JSP

Ex: - `<jsp:include page="header.jsp"/>`

2) <jsp:forward> forward request to another JSP

Ex: `<jsp:forward page="nextPage.jsp"/>`

3) <jsp:useBean> creates or retrieves a JavaBean object

Ex: `<jsp:useBean id="" class="" scope="" />`

4) <jsp:getProperty> Retrieves a property value from a JavaBean

Ex: `<jsp:getProperty name="" property="" />`

5) <jsp:setProperty>

6) <jsp:param> Parameter